

Mandatory Assignment 4

- **Deadline: October 30, 2023 by 23:59.**
 - The submission is in the form of a .pdf file which should be submitted through LearnIT.
 - We suggest that you either use LaTeX or scan hand-written notes to create .pdf.
 - You are encouraged to use the reference material while attempting the problems.
 - The mandatory assignment should be submitted individually. This does not mean that you cannot work in groups to solve these, but you must write up your solution in your own words and submit that.
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1. Give the PMF and CDF of $X \sim \text{DUnif}(1, 2, \dots, n)$ (Similar to first Problem from Set 6).
2. Expected value and linearity of expectation (Similar to Exercise 4.3 from Set 7).
 - (a) A fair die is rolled. Find the expected value of the roll.
 - (b) 2 fair dice are rolled. Find the expected total of the rolls.
 - (c) 500 fair dice are rolled. Find the expected total of the rolls.

1. Give the PMF and CDF of $X \sim \text{DUnif}(1, 2, \dots, n)$ (Similar to first Problem from Set 6).

$$X \sim \text{DUnif}(1, 2, \dots, n)$$

$$C = \{1, 2, \dots, n\}$$

$X \sim \text{DUnif}(C) \rightarrow$ r.v. X is distributed following discrete uniform distribution over values in C

$\hookrightarrow$ values can be listed/counted $\rightarrow$ Not intervals

$\hookrightarrow$ Probabilities of each value for x are the same

a) PMF

by definition: $P(X=x) = \frac{1}{\text{number of elements in set, that r.v. is defined in.}} = \frac{1}{|C|} = \frac{1}{n}$

$$f_X(x) = \frac{1}{n}$$

b) CDF

$$g_X(x) = \begin{cases} 0 & \text{if } x < 1 \\ \frac{x}{n} & \text{if } 0 < x < n \\ 1 & \text{if } x = n \end{cases}$$

and $\sum_{x=1}^n \frac{1}{n} = \lim \rightarrow 1 \Rightarrow = 1$ 1/2
 $\rightarrow x > 0 \Rightarrow$ Non Neg. 2/2

2. Expected value and linearity of expectation (Similar to Exercise 4.3 from Set 7).

(a) A fair die is rolled. Find the expected value of the roll.

(b) 2 fair dice are rolled. Find the expected total of the rolls.

(c) 500 fair dice are rolled. Find the expected total of the rolls.

by definition: expected value is the sum of individual values x times its probability

$$\sum_{j=1}^{\infty} x_j \cdot P(X=x_j)$$

a) Expected value of 6 sided dice

x	$P(X=x_j)$
1	$1/6$
2	$1/6$
3	$1/6$
4	$1/6$
5	$1/6$
6	$1/6$

$$\rightarrow E(x) = 1 \cdot (1/6) + 2 \cdot (1/6) + \dots + 6 \cdot (1/6)$$

$$= 21 \cdot (1/6)$$

$$E(x) = 3.5$$

b) $E(Y) = ?$

$Y = 2x$ fair dice rolled

$Y = 2x \cdot X$

by the rule $E(c \cdot X) = c \cdot E(x)$

$$E(2X) = 2 \cdot E(x)$$

$$= 2 \cdot 3.5$$

$$E(Y) = 7$$

c) Using $E(c \cdot X) = c \cdot E(x)$

$$E(500 \cdot X) = 500 \cdot 3.5$$

$$E(500x) = 1750$$